

# Forest Fire Simulation using NetLogo

CS 297 Final Exam  
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## Contents

|          |                                            |          |
|----------|--------------------------------------------|----------|
| <b>1</b> | <b>Fractal Frequency-Size Distribution</b> | <b>1</b> |
| <b>2</b> | <b>Immunizing a Forest</b>                 | <b>3</b> |
| <b>3</b> | <b>Self-organized Criticality</b>          | <b>5</b> |

## List of Figures

|   |                                                      |   |
|---|------------------------------------------------------|---|
| 1 | Forest fires exhibiting fractal properties. . . . .  | 2 |
| 2 | Forest fires with immunity. . . . .                  | 4 |
| 3 | Forest fires self-organizing critical state. . . . . | 6 |

## 1 Fractal Frequency-Size Distribution

According to the wiki, a double separation of time scales is necessary for the model to have a fractal:  $f \ll p \ll T_{max}$

In the netlogo simulation[1],  $p = 1$  since it is guaranteed that a tree will grow on an empty space. Therefore we need to set  $f$  small enough. Since the forest has large area, it is indeed  $p \ll T_{max}$  and fractal is observed. The screenshot below shows the simulation of a forest fire in a fractal manner.

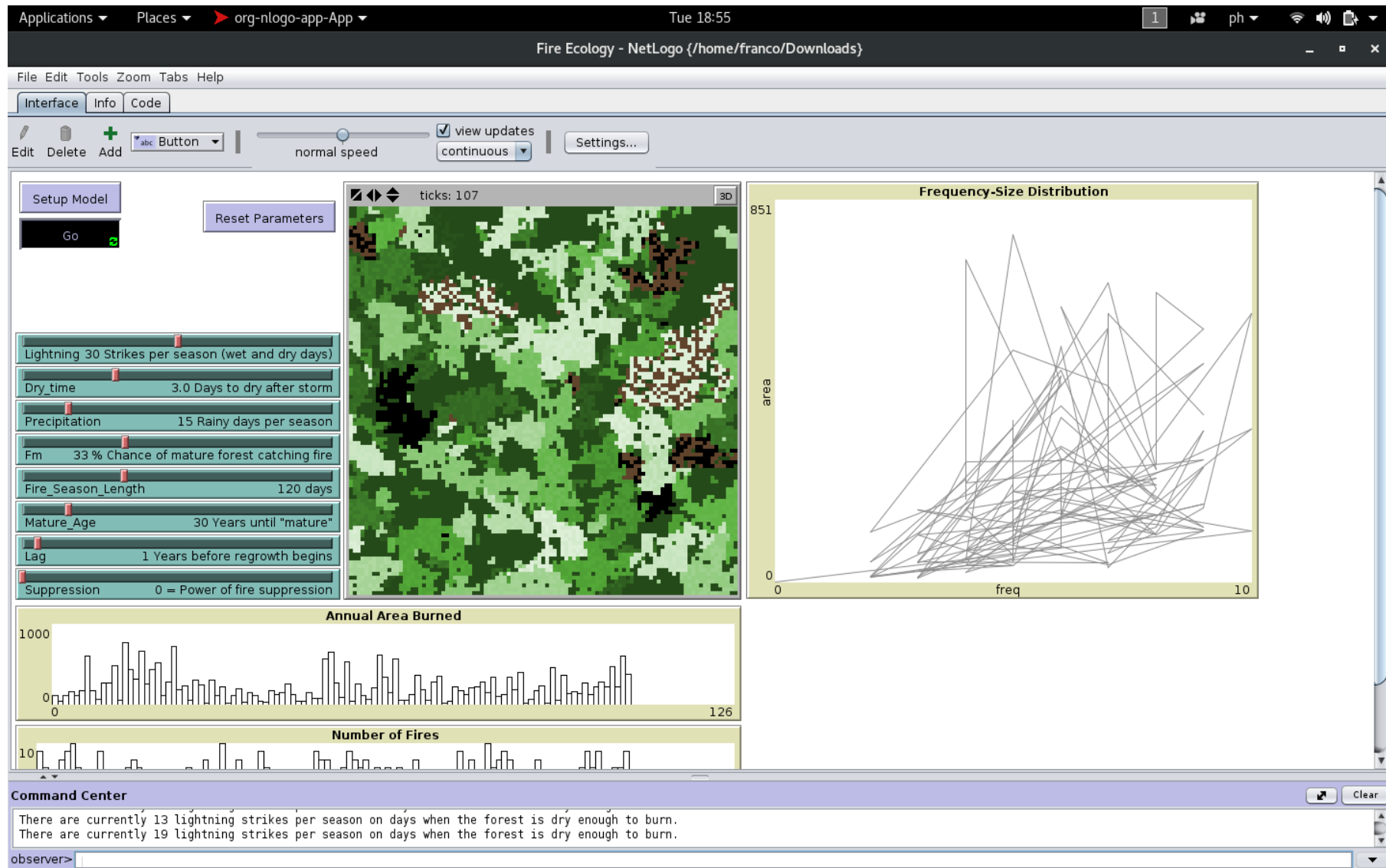


Figure 1: Forest fires exhibiting fractal properties.

The graphs of annual area burned per time is aperiodic and does not go into infinity, same as the number of fires per time. A graph has been added (on the right side) to plot the fire frequency to area trajectory at each time  $t$ . Consequently, the graph stays on a fixed region and does not approach infinity.

## 2 Immunizing a Forest

For the model to include an immunity dynamic introducing parameter  $g$  and modification of rule 5 in necessary such that "a green tree becomes a burning tree with probability  $(1 - g^n)$  if  $n$  of its nearest neighbors is burning"[2]. Otherwise it remains green. The parameter  $g$  introduces a critical threshold  $g^* = 1.2$  above and fires will not occur effectively. The following modifications are included in the source code:

- Adding a slider and a immunity parameter  $g$ . If  $g = 1$  then a tree is immune with probability 1.
- Modifying the boolean condition on the innermost **if** statement on the **spread** function to

```

- if age != 0 [
- set j sum[count fires-here] of neighbors4
- set p (1 - g ** j)
- if random-float 1 < p [ignite]
- ]

```

The figure below shows an 'immune' forest given  $g = 0.7$ . Annual area burned is relatively low. The large graph on the right shows trajectory of area to frequency as time increases. Notice that after some time it approaches a certain attractor meaning the annual area burned is kept bounded, thus the forest is fire-immune.

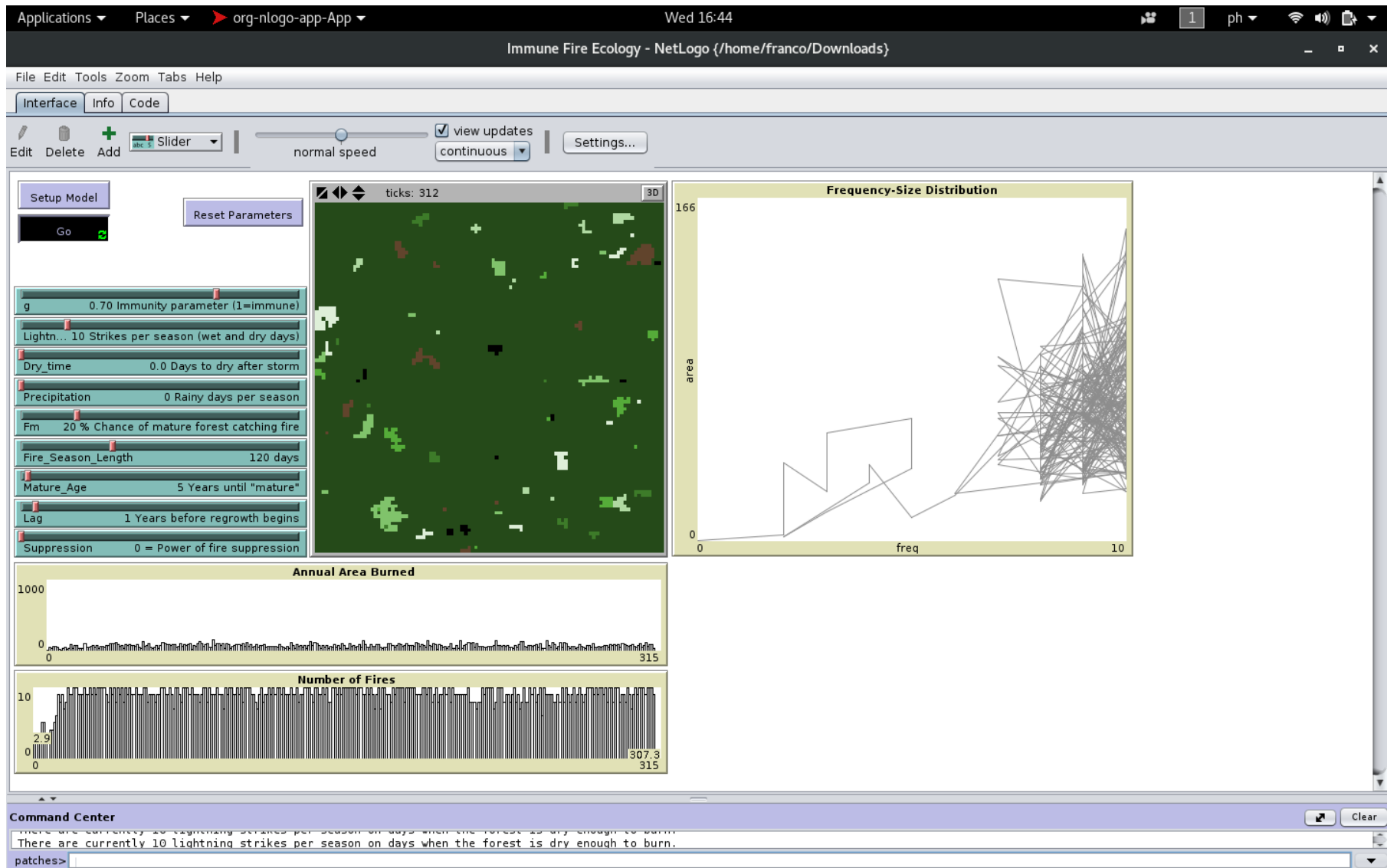


Figure 2: Forest fires with immunity.

### 3 Self-organized Criticality

From wiki, self-organized criticality is a property of dynamical systems where the critical or fixed point is an attractor. In the paper of [3], a forest-fire leads to a self-organized critical state in the limit  $f \rightarrow 0$  when the times scales of tree growth and burning down of clusters are separated.

In letting  $f \rightarrow 0$  the number of lightning strikes and  $f$  has set to a low value so that the time between two consecutive strikes per cell is very large. Mature age is set to 1 year before growth begins. In this manner, most of the forest will burn and mean density will be kept low as possible as in a critical state.

Again, the double separation of time scales has been observed in the simulation. In the netlogo simulation[1] parameters are set as follows in the screenshot below.

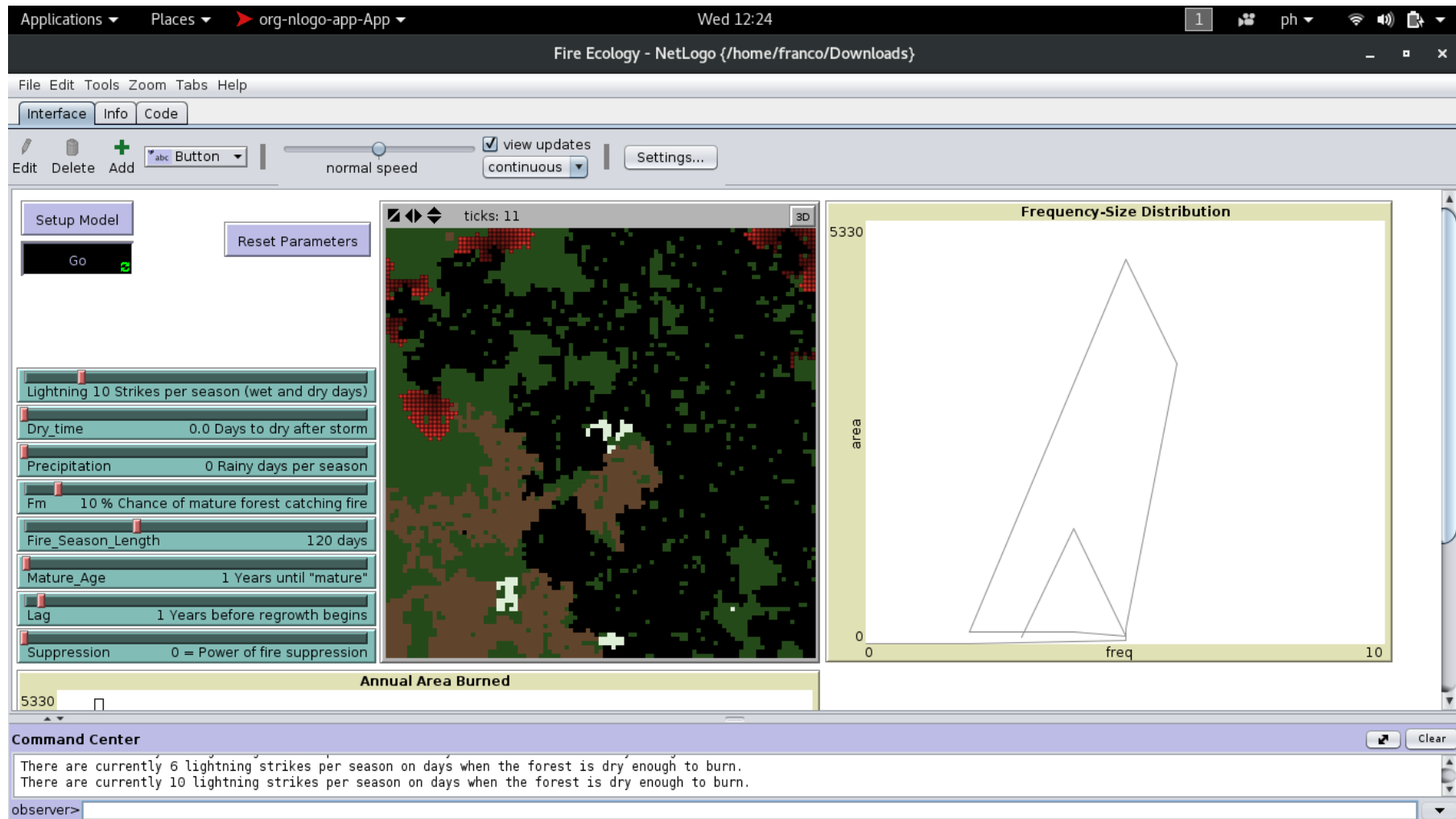


Figure 3: Forest fires self-organizing critical state.

## References

- [1] R. Kelly, (2009). *NetLogo Fire Ecology model*. Department of Plant Biology, University of Illinois, Urbana, IL. <http://ccl.northwestern.edu/netlogo/models/community/Fire%20Ecology>
- [2] M. Yoder, L. Turcotte, J. Rundle, (2011). *Forest-fire model with natural fire resistance*. Department of Physics, University of California Davis.
- [3] B. Drossel and F. Schwabl (1992). *Self-Organized Critical Forest-Fire Model*.